

SM PHARMACEUTICALS SDN. BHD.

FUSIX INJECTION

(FRUSEMIDE INJECTION 20 MG/2ML)

Description:

A colourless clear liquid, free from foreign impurity and particulate matter.

Composition:

Contains: Frusemide 20 mg/2ml

Actions and mode or mechanisms of action:

The pharmacologic effects of frusemide are similar to those of ethacrynic acid. The exact mode of action frusemide has not been clearly defined. In contrast to ethacrynic acid, it does not bind sulfhydryl groups of renal cellular proteins. Frusemide inhibits the reabsorption of electrolytes in the ascending limb of the loop of Henle. The drug also decreases reabsorption of sodium and chloride and increases potassium excretion in the distal renal tubule and exerts a direct effect on electrolyte transport at the proximal tubule. Frusemide does not inhibit carbonic anhydrase and is not an aldosterone antagonist.

Frusemide diuresis results in enhanced excretion of sodium, chloride, potassium, hydrogen, calcium, magnesium, ammonium, bicarbonate, and possibly phosphate. Chloride excretion exceeds that of sodium. In studies in patients with normal renal function, the diuretic response was similar following oral or IV administration of equal doses of frusemide than after equal oral doses. Excessive losses of potassium, hydrogen, and chloride may result in metabolic alkalosis. Urinary pH usually falls after administration of frusemide. However, increased bicarbonate excretion in some patients may temporarily raise urinary pH. Low doses of frusemide promote uric acid retention while large IV doses may cause temporary uricosuria. Maximum diuresis and electrolyte loss is greater with frusemide than with the thiazides or most other diuretics except ethacrynic acid. Like the thiazide diuretics and ethacrynic acid, the effectiveness of frusemide is independent of the acid-base balance of the patient.

Frusemide has some renal vasodilator effect. Renal vascular resistance decreases and renal blood flow increases following administration of the drug. When large doses of frusemide are administered to patients with chronic renal insufficiency, glomerular filtration rate may be increased temporarily. A fall in renal blood flow and glomerular filtration rate may occur if excessive drug-induced diuresis results in a reduction in plasma volume.

As with other diuretics, a hypotension effect may result from decreased plasma volume in patients receiving frusemide. However, the drug has been reported to produce only mild decreases in the supine systolic blood pressure and in the erect systolic and diastolic blood pressures when administered alone in the recommended oral dosage.

Frusemide appears to have less effect on carbohydrate metabolism and blood glucose concentrations than do the thiazides. However, the drug may cause elevations of blood glucose, glycosuria, and alterations in glucose tolerance possibly as a result of hypokalemia.

Pharmacology:

Absorption

In one study in patients with normal renal function, approximately 60% of a single 80-mg oral dose of frusemide was absorbed from GI tract. When administered to fasting adults in this dosage, the drug appeared in the serum within 10 minutes, reached a peak concentration of 2.3 $\mu\text{g} / \text{ml}$ in 60-70 minutes, and was almost completely cleared from the serum in 4 hours. When the same dose was given after a meal, the serum concentration of frusemide increased slowly to a peak of about 1 $\mu\text{g} / \text{ml}$ after 2 hours and similar concentrations were present 4 hours after ingestion. However, a similar diuretic response occurred regardless of whether the drug was given with food or to fasting patients. In another study, the rate and extent of absorption varied considerably when 1 g of frusemide was given orally to uremic patients. An average of 76% of a dose was absorbed, and peak plasma concentrations were achieved within 2-9 hours (average 4.4 hours). Serum concentrations required to produce maximum diuresis are not known, and it has been reported that the magnitude of response does not correlate with either the peak or the mean serum concentrations.

The diuretic effect of orally administered frusemide is apparent within 30 minutes to 1 hour and is maximal in the first or second hour. The duration of action is usually 6-8 hours. The maximum hypotension effect may not be apparent until several days after frusemide therapy is begun. In patients with severely impaired renal function, the diuretic response may be prolonged.

After IV administration of frusemide, diuresis occurs within 5 minutes, reaches a maximum within 20-60 minutes, and persists for approximately 2 hours. After IM administration, peak plasma concentrations are attained within 30 minutes. Onset of diuresis occurs somewhat later than after IV administration. In patients with severely impaired renal function, the diuretic response may be prolonged.

Distribution

Only limited information is available on the distribution of frusemide. The drug crosses the placenta and is distributed into milk.

Frusemide is approximately 95% bound to plasma proteins in both normal and azotemic patients.

Elimination

Plasma concentrations of frusemide decline in a biphasic manner. Various investigators have reported a wide range of elimination half-lives for frusemide. The elimination half-life was more prolonged in 1 patient with concomitant liver disease.

In patients with normal renal function, a small amount of frusemide is metabolized in the liver to the defurfurilated derivative, 4-chloro-5-sulfamoylanthranilic acid. Frusemide and its metabolite are rapidly excreted in urine by glomerular filtration and by secretion from the proximal tubule. In patients with normal renal function, approximately 50% of an oral dose is excreted in the urine within 24 hours. 69-97% of these amounts is excreted in the first 4 hours. The remainder of the drug is eliminated by nonrenal mechanisms including degradation in the liver and excretion of unchanged drug in the feces. In patients with marked renal impairment without liver disease, nonrenal clearance of frusemide is increased so that up to 98% of the drug is removed from the plasma within 24 hours. Frusemide is not removed by hemodialysis.

Indications:

Frusemide is indicated in the treatment of edema associated with congestive heart failure, hepatic cirrhosis and renal disease (including nephrotic syndrome). It is also indicated as an adjunct in the treatment of acute pulmonary edema and hypertensive crisis.

Contraindications:

Frusemide is contraindicated in patients with anuria. The drug is contraindicated for further use if increasing azotemia and/or oliguria occur during the treatment of severe, progressive renal disease. In patients with hepatic coma or electrolyte depletion, therapy should not be instituted until the basic condition is improved or corrected. Frusemide is also contraindicated in patients with a history of hypersensitivity to the drug and to sulphonamides.

Side Effects / Adverse Reactions:

The most common side-effect associated with frusemide therapy is fluid and electrolyte imbalance including hyponatraemia, hypokalaemia, and hypochloraemic alkalosis,

particularly after large doses or prolonged administration. Unlike the thiazide diuretics, it increases the urinary excretion of calcium. Nephrocalcinosis has been reported when frusemide has been used to treat preterm infants. Frusemide may provoke hyperglycaemia and glycosuria, but probably to a lesser extent than the thiazide diuretics. It may cause hyperuricaemia and precipitate attacks of gout in some patients. Other side-effects are relatively uncommon, and include gastro-intestinal disturbances, blurred vision, yellow vision, dizziness, headache, and orthostatic hypotension. Skin rashes and photosensitivity reactions, although rare, may be severe and hypersensitivity reactions including interstitial nephritis occur rarely. Pancreatitis is more common at high doses and cholestatic jaundice has been reported. Bone marrow depression may occur rarely : agranulocytosis, thrombocytopenia, and leucopenia have been reported. Tinnitus and deafness may rarely occur in particular during rapid high dose parenteral frusemide therapy. Rarely, deafness may be permanent particularly if frusemide has been given to patients taking other ototoxic drugs.

Diuresis (strong) - The most commonly encountered adverse effects attributable to frusemide are fluid depletion and electrolyte disturbance arising from frusemide's diuretic actions.

Levels of serum lipids - Most studies into the effects of diuretics on bloods-lipids concentrations have used thiazides and long-term studies have suggested that adverse effects on blood lipids may be transitory, although the complete lipid-lipoprotein analysis necessary for full evaluation of the effects has generally not been preformed. The few studies into the effects of frusemide suggest that it may adversely influence blood-lipid concentrations during short-term administration.

Adverse dermatologic and / or hypersensitivity reactions to frusemide include purpura, photosensitivity, rash, urticaria, pruritus, exfoliative dermatitis, erythema multiforme, and necrotizing angitis (vasculitis, cutaneous vasculitis). Patients with known sulfonamide sensitivity may show allergic reactions to frusemide. Anaphylaxis, manifested as urticaria, angiodema, and hypotension, occurred with 5 minutes after IV administration of frusemide in at least one patients; subsequent intradermal skin testing showed sensitivity to frusemide and other sulfonamides.

Precautions / Warnings:

Frusemide's precautions and contra-indications which are dependent on its effects on fluid and electrolyte balance are similar to those of the thiazide diuretics. Although frusemide is used in high doses for oliguria due to chronic or acute renal insufficiency it should not be given in anuria or in renal failure due to nephrotoxic or hepatotoxic drugs nor in renal failure associated with hepatic coma. Frusemide should not be given in pre-comatose states associated with hepatic cirrhosis. It should be used with care in

patients with prostatic hypertrophy or impairment of micturition since it can precipitate acute urinary retention.

Frusemide should be used with caution in pregnancy.

Cross sensitivity and/or related problems

Patients sensitive to sulfonamides (including thiazide diuretics) may be sensitive to Frusemide also.

Mutagenicity

Mutagenicity studies have not been conducted.

Pediatrics

Caution is required in neonates because of the prolonged half-life of Frusemide.

Usual pediatrics doses may be used, but dosing interval should be extended.

Geriatrics

Although appropriate studies on the relationship of age to the effects of loop diuretics have not been performed in the geriatric population, the elderly may be more sensitive to the hypertensive and electrolytes effects. In addition, elderly patients are at greater risk of developing circulatory collapse and thromboembolic episodes. Elderly patients are also more likely to have age-related renal function impairment, which may require adjustment of dosage or dosing interval in patients receiving loop diuretics.

Pregnancy/Reproduction

Pregnancy – Pregnant women should be advised to contact physician before taking this medication, since routine use of diuretics during normal pregnancy is inappropriate and exposes mother and fetus to unnecessary hazard. Diuretics do not prevent development of toxemia of pregnancy, and there is no satisfactory evidence that they are useful in the treatment of toxemia. Diuretics are indicated only in the treatment of edema due to pathologic causes or as a short cause of treatment in patients with severe hypervolemia.

Frusemide crosses the placenta. Studies in humans have not been done. Studies in rabbits and mice have shown that it causes an increased incidence of hydronephrosis in the fetus. In rabbits, unexplained maternal deaths and abortions have occurred at doses 2 to 8 times the maximum recommended human dose. FDA Pregnancy Category C.

Breast-feeding

It is distributed into breast-milk.

Drug Interactions

Frusemide may enhance the nephrotoxicity of cephalosporin antibiotics such as cephalothin. It can enhance the ototoxicity of aminoglycoside antibiotics. Like other diuretics, frusemide enhances the hypotensive action of antihypertensive drugs. Particular care should be taken with ACE inhibitors since combination with frusemide can result in marked reduction in blood pressure.

Recommended Dosage, Dosage Schedule and Route of Administration:

In an emergency or when oral therapy can not be given, the equivalent of 20 to 50 mg of frusemide may be administered by intramuscular or slow intravenous injection as the sodium salt at a rate not exceeding 4 mg per minute. If necessary further doses may be given, increasing by 20 mg increments and not given more often than every 2 hours. If doses greater than 50 mg are required it is recommended that they be given by slow intravenous infusion. For pulmonary oedema, sources in the USA have recommended that if an initial slow intravenous injection of 40 mg (over 1 to 2 minutes) does not produce a satisfactory response within one hour, the dose may be increased to 80 mg given slowly intravenously.

For children, suggested doses by injection are 0.5 to 1.5 mg per kg daily up to a maximum of 20 mg daily, although doses of up to 6 mg per kg have been suggested. In the treatment of hypertension, frusemide is given in doses of 40 to 80 mg daily, either alone, or in conjunction with other antihypertensive agents. In susceptible patients potassium supplements or potassium-sparing diuretics may be necessary, but see hydrochlorothiazide.

High-dose therapy. In the management of oliguria in acute or chronic renal failure where the glomerular filtration rate is less than 20 ml per minute frusemide 250 mg diluted to 250 ml in a suitable diluent is infused over one hour. If urine output is insufficient within the next hour, this dose may be followed by 500 mg added to an appropriate infusion fluid, the total volume of which must be governed by the patient's state of hydration, and infused over approximately 2 hours. If a satisfactory urine output has still not been achieved within one hour of the end of the second infusion then a third dose of 1 g may be infused over approximately 4 hours. The rate of infusion should never exceed 4 mg per minute. In oliguric or anuric patients with significant fluid overload, the injection may be given without dilution directly into the vein, using a constant-rate infusion pump with a micrometer screw-gauge adjustment; the rate of administration should still never exceed 4 mg per minute.

Patients who do not respond to a dose of 1 g probably require dialysis. If the response to either method of administration is satisfactory, the effective dose (of up to 1 g) may then be repeated every 24 hours. Dosage adjustments should subsequently be made according to the patient's response. Alternatively, treatment may be maintained by mouth; 500 mg should be given by mouth for each 250 mg required by injection.

During treatment with these high-dose forms of frusemide therapy, careful laboratory control is essential. Fluid balance and electrolytes should be carefully controlled and, in particular, in patients with shock, measures should be taken to correct the blood pressure and circulating blood volume, before commencing this type of treatment. High dose frusemide therapy is contra-indicated in renal failure caused by nephrotoxic or hepatotoxic agents, and in renal failure associated with hepatic coma.

Symptoms and Treatment for Overdose and Antidote(s):

Overdose will lead to fluid and salt depletion subsequent to initial excessive diuresis. Medical help must be called in without delay so as to ensure immediate measures for an adequate restoration of water and electrolytes balances.

Storage Conditions, User Instructions and Pharmaceutical Precautions:

Store below 30°C.

Protect from light and freezing.

Do not use if solution is yellow.

Frusamide injection is a mildly buffered alkaline solution and should not be mixed with highly acidic solutions.

Packing / Pack Sizes:

2 ml ampoules, 10 ampoules in a carton

Shelf life:

3 years

Manufacturer and Product Registration Holder:

SM PHARMACEUTICALS SDN BHD (218620-M)

Lot 88, Sungai Petani Industrial Estate

08000 Sungai Petani,

Kedah Darul Aman,

Malaysia.

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